

PPA-5: Physical and Educational Spatial Computing Devices

Title

Physical and Educational Devices for Demonstrating and Implementing Symmetry-Based Computation

Field of the Invention

The invention relates to physical devices, kits, and educational systems that embody symmetry-based computation in tangible form.

Background

Abstract computation is difficult to understand intuitively. Existing educational tools rely on symbolic representations that obscure spatial logic.

There is a need for physical devices that:

- embody computation spatially,
 - allow direct manipulation,
 - and demonstrate symmetry-constrained logic.
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Summary

The invention discloses physical devices comprising:

- polyhedral components,
- discrete state elements,
- and mechanisms for symmetry-constrained transitions.

Such devices may be used for:

- education,
- research,

- demonstration,
 - or physical computation.
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Detailed Description

Physical Embodiments

Devices may include:

- polyhedral elements,
 - rotational connectors,
 - constraint mechanisms,
 - visual or tactile indicators of state.
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Educational Systems

Kits may include:

- multiple polyhedral logic points,
 - guides for exploration,
 - exercises demonstrating computation by symmetry.
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Advantages

- Tangible computation
 - Intuitive learning
 - Demonstrates non-Boolean logic
 - Supports commercialization
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Provisional Statement

This application claims priority for physical and educational devices implementing symmetry-based computation.